

**Semipalmated Sandpiper (SESA) US Atlantic Coast Migration Stopover Site Analysis
using eBird Citizen Science Data (2001-2025)**

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Abstract

The Semipalmated sandpiper (SESA) *Calidris pusilla* is an IUCN Near Threatened migratory shorebird that is also classified as an orange alert tipping point species due to its accelerated population declines occurring over the last decade. Threats to migratory stopover sites along the Atlantic Coast of the United States include habitat loss, changes in land cover, and coastal development. The objective of this study is to determine if land cover class and habitat characteristics can be used to predict SESA presence at stopover sites to determine their usage of these locations for refueling and resting along their migratory journeys, and to assess whether or not detection rates at these stopover sites are changing over the course of the study period. Approximately six million eBird checklists were filtered to the 10 states along the Atlantic Coast that were focused on in this study during the migration period from the years of 2001-2025. 1,495 sites were used in modeling, 499 presence sites along with 996 absence sites. Predictor variables included NLCD land cover classes, elevation of the site, distance of the site from the coast, as well as climate variables such as temperature, wind, and precipitation. A Logistic Regression (LR) model (baseline model) was compared to a Boosted Regression Trees (BRT) model. Both models were evaluated using AUC-ROC, Delong's Test, and 5 fold cross validation. BRT outperformed LR, supported by an AUC of 0.819 compared to LR AUC of 0.783, with a p value = 0.0076 showing statistically significant performance. Physical habitat characteristics including elevation, distance to coast, and temperature dominated predictions and accounted for ~65% of predictive power within the model. Wetland land cover classification accounted for 5.77% of predictive power and was the most important land cover class feature. Detection rates were stable among the 500 presence sites assessed, and only one site, L455080 in Rhode Island, showed a decline in detection that was statistically significant. The hypothesis was partially supported, as wetland as the habitat type does predict presence of SESA, however physical characteristics are much more significant when analyzing feature importance. Stability that is seen in the top 500 sites suggests that the declines could be occurring in the less popular stopover site locations, which are not focused on in this analysis.

Introduction

The Semipalmated Sandpiper (SESA) *Calidris pusilla* is a small migratory shorebird whose migration range consists of a large portion of North America, including the Atlantic Coast of the United States. They breed in the Arctic tundra of northern Canada and Alaska in late spring and early summer, and make their way down to South America for the winter. According to the IUCN Red List, they are listed as Near Threatened as a species (BirdLife International, 2024) . They are also classified as an orange alert tipping point species, which is defined as birds showing long-term population losses and accelerated declines within the past decade (Partners in Flight, 2022) .

Stopover sites, areas where migratory birds rest and refuel during their migration, are essential to the success of migratory birds such as SESA. There are many threats to these stopover sites, primarily including habitat loss, land cover changes, and coastal development. This analysis aims to add to the studies that have previously been done predicting stopover site usage with quantitative models using eBird citizen science data (Sullivan et al., 2009).

There were two main research questions this analysis aims to answer:

1. What habitat and land cover characteristics predict SESA stopover site usage?
2. Have detection rates at stopover sites changed over time?

The hypothesis states that SESA stopover site usage is predictable using land cover characteristics, and sites that have experienced habitat degradation during the years of study are less likely to be used as stopover sites and therefore have a lower chance of SESA being present.

Data & Feature Engineering

The focal species of this analysis, SESA, are of particular concern during the migration season which ranges from May to September. The geographic range focused on are states along the Atlantic Coast of the US. The Atlantic Coast is defined as the following states: Connecticut, Delaware, Maine, Maryland, Massachusetts, New Hampshire, New Jersey, New York, Rhode Island, and Virginia.

Data was obtained from eBird through the Cornell Lab of Ornithology and was filtered to the specific study area and migration months during the years 2001-2025. There were ~12 million total checklists in the ebird sampling file, with ~6 million completed checklists in the sampling file. Figure 1 shows detections by season along the US Atlantic Coast.

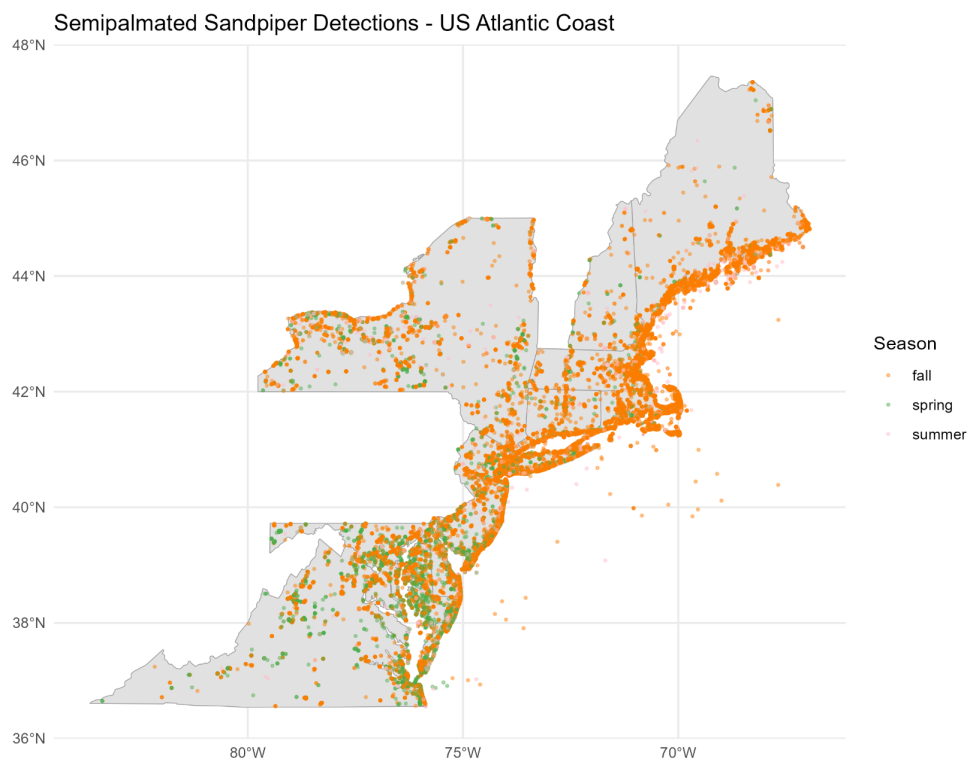


Figure 1. SESA detection by season

Completed checklists were used so that both presence and absence sites could be identified using checklists. The data was then aggregated to unique “locality_id” which resulted in 23,076 unique presence locations (locations where SESA was detected). From this compiled list of detection sites the top 500 most used sites were identified and extracted to use for modeling. The top 500 sites were determined based on the checklist frequency of each site. Absences were randomly sampled from completed checklists that did not have any SESA detections, and 1,000 locations were extracted to use in modeling. A 1:2 presence to absence ratio was used due to proven success in prior models with higher absence sites than presence sites (Barbet-Massin et al., 2012). As seen in figure 2, the final dataset consisted of 1495 locations due to 5 rows being dropped from the dataset because of missing land cover class data.



Figure 2. Presence absence class distribution

Data used as predictor variables was collected from multiple accessible online sources as well as preexisting R packages. National Land Cover Database (NLCD) land cover data from 2001 and 2025 was used to compare the type of habitat at each location and determine whether there was habitat change as a whole and whether that change was positive or negative. Elevation data was obtained from the R package “elevatr”. Distance to the coastline in kilometers (km) was obtained via the R package “NaturalEarth”. Climate data, such as wind speed, temperature, precipitation was obtained from the R package “nasapower”. A few features were engineered by comparing NLCD data from 2001 and 2025: land cover changed, habitat degraded, and habitat improved.

One hot encoding (OHE) was performed on NLCD features, and the first dummy was removed to ensure that there is no multicollinearity. The reference feature was “Open Water”, this means that all of the other land class features are interpreted relative to that feature in the models. Low variance features and leaky features were removed. The only low variance feature was “ShrubScrub”, and leaky features included “locality_id” and “n_checklists”, so not many features were removed. The train test split used was 70/30, set.seed(42) was used for reproducibility.

Exploratory data analysis was done to get a sense of the data used. Land cover at presence sites is dominated by four classes as shown in Figure 3, developed low intensity, emergent herbaceous wetlands, open water, and developed open space. 13.1% of sites experienced a land cover change, either positive or negative, from 2001-2025. As shown in figure 4, features did not have

any severe multicollinearity after duplicate 2001 features were removed once habitat change columns were created.

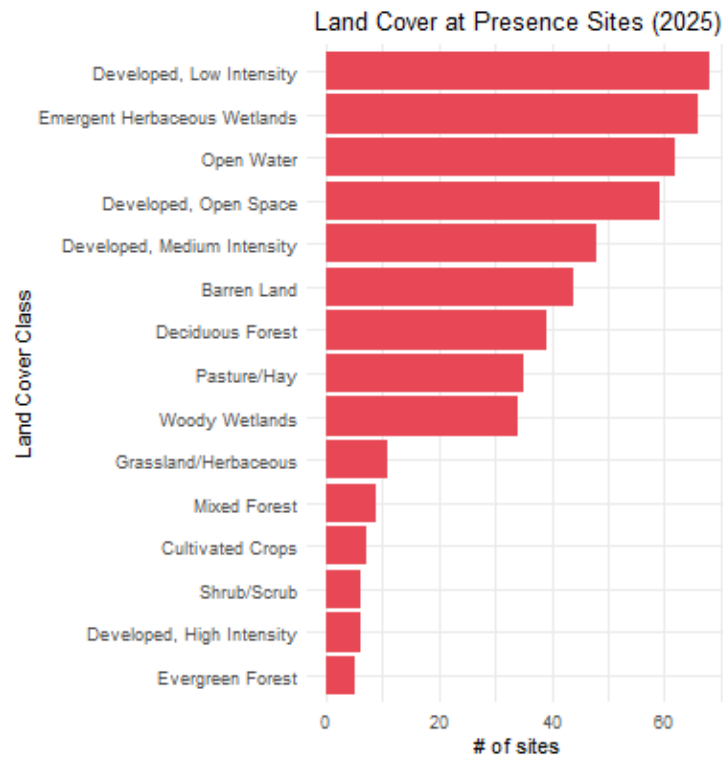


Figure 3. Land cover at presence sites

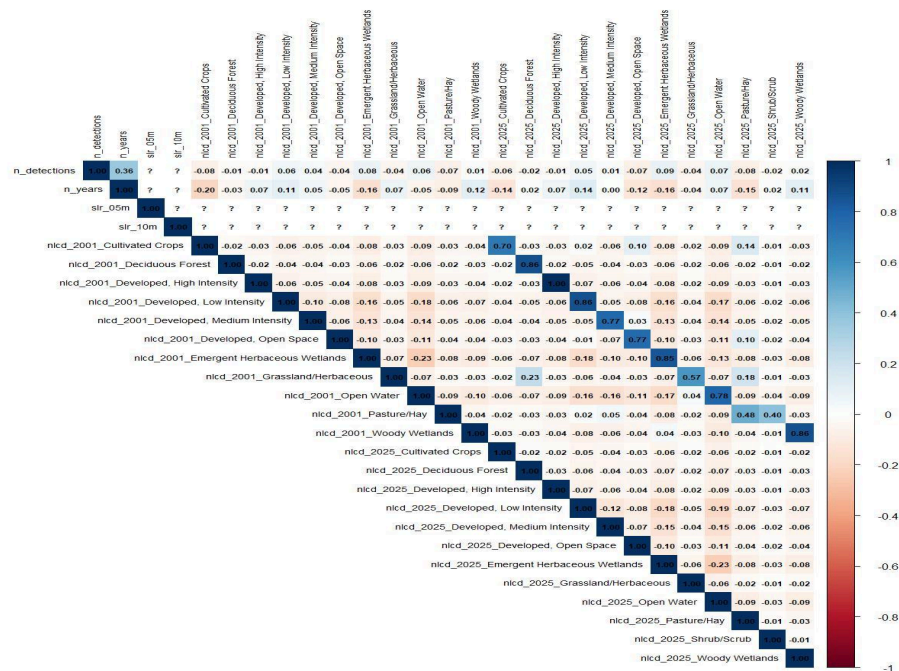


Figure 4. Feature correlation heatmap

Methodology

Two models were compared in this classification problem analysis, Logistic Regression (LR) and Boosted Regression Trees (BRT). LR was the baseline model, used to estimate the probability of the presence of SESA at a stopover site. The function for the LR model is defined as

$$P(y=1) = 1 / (1 + e^{-z})$$

where z is the linear combination of the predictor variables, calculated as

$$z = B_0 + B_1x_1 + B_2x_2 + \dots + B_nx_n$$

where B_0 is the intercept and B_1 and B_2 are the coefficients for each predictor variable (x_1, x_2) which includes features such as elevation, weather data, land cover classification, etc. The output of the function is a probability 0-1 which gets classified as a presence or absence site based on the decision threshold (0.4). A threshold of 0.4 was decided on to improve recall which in turn increases the importance of flagging presence sites correctly as it is more imperative to be cautious about identifying potential stopover sites.

The BRT model was then used to compare performance of a more complex model to the baseline model. BRT used a bernoulli distribution with 1000 trees (optimal number of trees (882) was selected using 5-fold cross validation) and an interaction.depth of 3 and a shrinkage (learning rate) of 0.01. The function for the BRT model, which builds an ensemble of decision trees, is defined as

$$F(x) = F_0 + \sum_t \gamma_t h(x; a_t)$$

where F_0 is the initial prediction, $h(x; a_t)$ is a weak decision tree fit at iteration t , and γ_t is the learning rate. Each tree is fit to the residual errors of the previous iteration, this allows the model to iteratively improve its predictions.

Hyperparameter tuning was completed using 5-fold cross validation to select the optimal number of trees for BRT, 882, the test AUC improved from 0.811 to 0.819 once the model was tuned as shown in table 1.

	Untuned BRT	Tuned BRT
n.trees	100	882
interaction.depth	1	3
shrinkage	0.1	0.01
Test AUC	0.811	0.819

Table 1: BRT, BRT Tuned

Analysis was conducted using R via RStudio. The base R function `glm()` was used to fit the logistic regression model and the boosted regression model was fit using the `gbm` package. One hot encoding was done via the `fastDummies` package, and evaluation metrics and statistical testing were done via the `pROC` package. Additional data was acquired via `elevatr`, `rnaturalearth` and `nasapower` as mentioned in earlier sections.

Both models were evaluated on the test set (30% of the full dataset). Evaluation metrics include accuracy, AUC-ROC, precision, recall, and F1 score. The main metric focused on was the AUC-ROC, which measures how well the model tells the difference between presence and absence sites. AUC-ROC = 0.5 is equivalent to a 50/50 chance of correctly predicting presence and absence, and AUC-ROC = 1.0 is a perfect separation of presence and absence. Both LR and BRT models fell in the range above 0.5, meaning their performance was meaningful in

separating presence and absence sites. The Delong method was used to calculate 95% confidence intervals (CI) for AUC. 5 fold cross validation (CV) AUC was calculated and used to ensure generalizability, similar values for test AUC and CV AUC prove the model is not overfitting the training data. Delong's test was used to assess the AUC values of LR and BRT for statistical significance. Delong's was used in place of a t-test because it is tailored for comparison of AUC values of models that are evaluated on the same test set as each other. A confusion matrix was used to compare the predictions of each model and analyze the strengths and weaknesses of each model using true positive, false positive, true negative, and false negative. Furthermore, error analysis was done to determine if there were patterns in incorrect predictions, particularly false negatives and false positives. Finally learning curves were produced to visualize the model convergence with increased training size.

Temporal analysis was conducted with the purpose of assessing if usage of presence sites by SESA has changed over time. Detections were normalized to control for the increase in ebird usage over time, so detection rate was calculated as detections per checklist. Both aggregated and site trends were analyzed. Aggregate trends assess the overall detection rates of SESA at presence sites, site level trends were analyzed using linear regression to highlight those that have statistically significant declines in detection rate.

Results

The dataset consists of 1,495 unique locations, comprising 499 presence sites and 996 absence sites. Top land cover classes at presence sites include developed open space, low intensity

development, and emergent herbaceous wetlands. 13.1% of sites experienced land cover change during the study period.

The baseline LR model performed adequately on the data, but missed some patterns due to its simplicity. LR had an AUC score of 0.783, with a 95% CI between 0.740 - 0.825 and a CV AUC of 0.777. Accuracy of the LR model was 73.1%, while precision and recall were lower (0.600 and 0.643 respectively) resulting in an F1 score of 0.621. Specificity of the LR model was 0.776.

The secondary BRT model had better performance than the baseline model. BRT had an AUC score of 0.819, with a 95% CI between 0.779 - 0.859 and a CV AUC of 0.803. Accuracy of the BRT model was 76.6%, while precision and recall were slightly lower (0.706 and 0.545 respectively) resulting in an F1 score of 0.615. Specificity of the BRT model was 0.881, and the optimal number of trees was 882.

Table 2 shows a comparison of the baseline logistic regression model with the boosted regression trees model, compared to the naive baseline.

Metric	Naive Baseline	Logistic Regression	Boosted Regression Trees
AUC	0.500	0.783	0.819
CV AUC	-	0.777	0.803
Accuracy	67.7%	73.1	76.6
Precision	-	0.600	0.706
Recall	-	0.643	0.545
F1	-	0.621	0.615

Table 2: Model Evaluation Metrics Comparison

By comparing the confusion matrix of each model the strengths and weaknesses can be identified (Figure 5 and 6). Both models were strong at flagging absence sites, LR identified 229 absence sites correctly with 55 false negatives and BRT identified 260 absence sites correctly with 70 false negatives. Both models performed slightly differently when identifying presence sites. LR correctly identified 99 presence sites while having 66 false positives, where BRT correctly identified 84 presence sites while having 35 false positives.

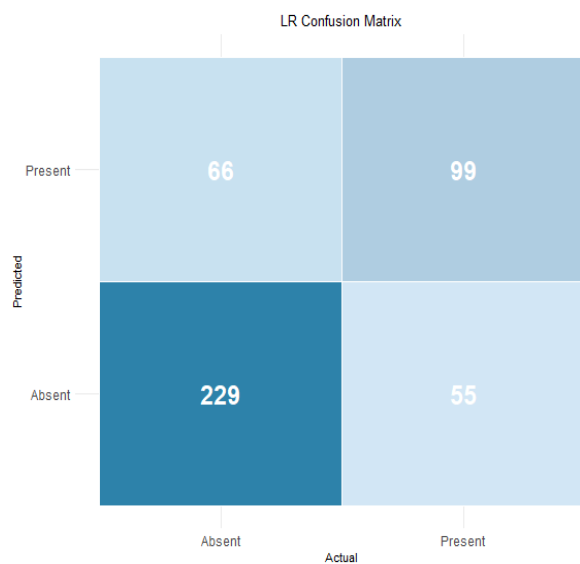


Figure 5. LR confusion matrix

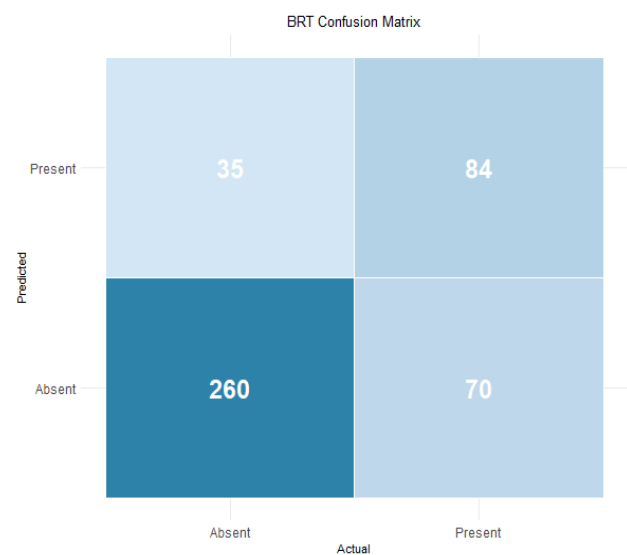


Figure 6. BRT confusion matrix

A ROC curve was utilized to visualize the performance of both models against a naive baseline. By plotting the trade off between sensitivity (the true positive rate) and specificity (the false positive rate) at every decision threshold the plot shows how AUC values for each model outperform the naive baseline.

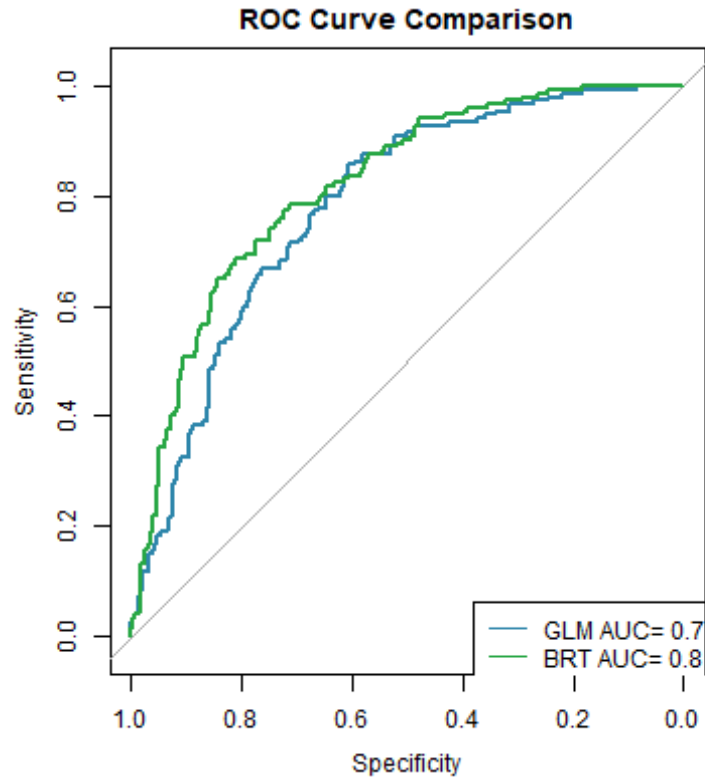


Figure 7. LR and BRT model ROC curve comparison

Delong's test was used to determine statistical significance between the two models because it accounts for correlated AUC scores when two models are tested on the same test. Results include $z = -2.6696$, $p\text{-value} = 0.0076$, 95% CI -0.063 to -0.010. $P\text{-value} < 0.05$ determines statistical significance between the two models AUC scores. The confidence interval doesn't include 0 which indicates that these two models do not perform equally, and the negative values of the confidence interval indicate that BRT outperforms LR. The cross validation (CV) AUC, LR: 0.777 and BRT: 0.803, shows the models are stable regardless of which subset they were trained on.

When comparing the learning curves the convergence differences in the models is noticeable, as shown in figures 8 and 9. LR begins with higher train and lower validation AUC scores and ends in a convergence that signifies that there is no underfitting or overfitting in the model. BRT also

begins with higher train and lower validation AUC, but does not converge. Instead there is still a noticeable gap between the train and validation AUC scores, which indicates potential overfitting or the need for additional data or features to allow for the model to converge.

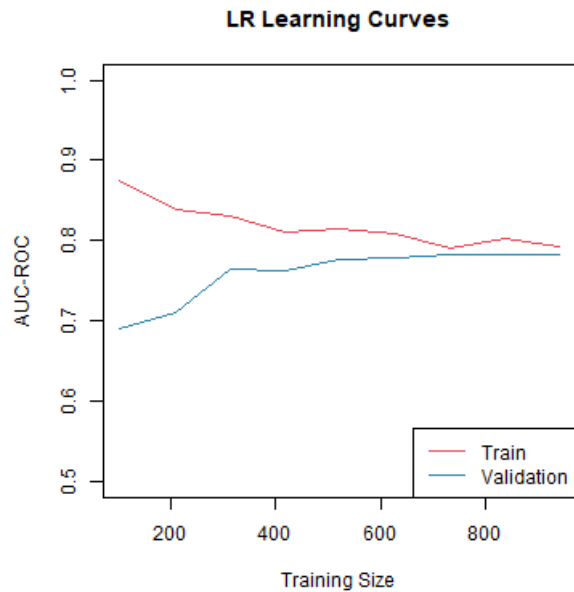


Figure 8. LR learning curve

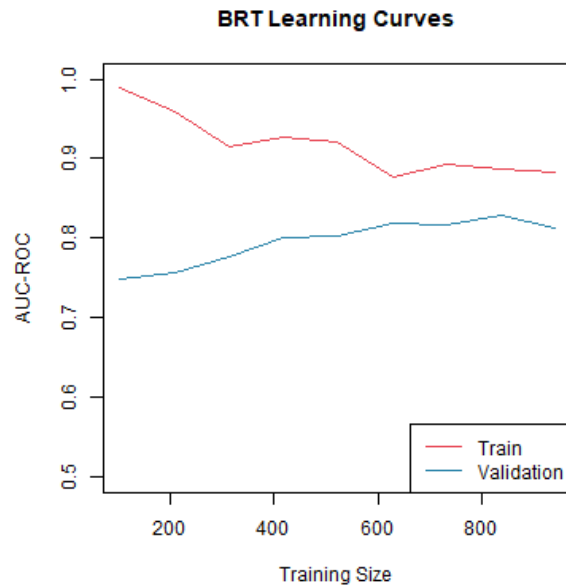


Figure 9. BRT learning curve

Feature importance analysis revealed that most of the predictive power in the model came from three features, none of which were land cover data. These three features were elevation (24.97%), distance to coast (23.97%), and temperature (15%). The land cover feature with the highest relative influence was emergent herbaceous wetlands (5.77%).

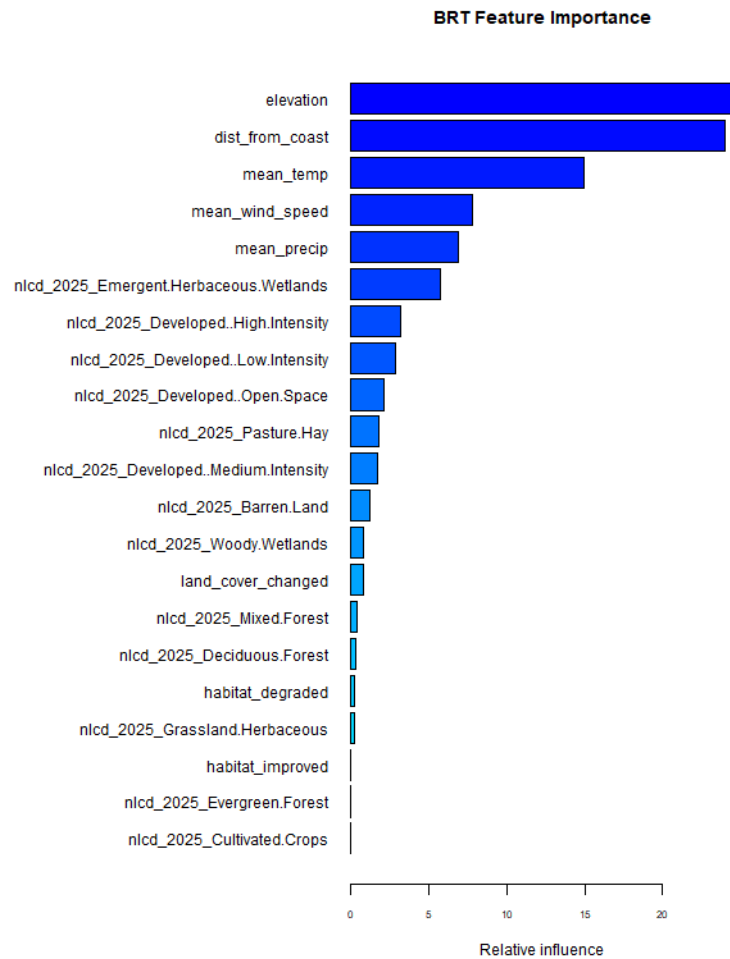


Figure 10. BRT Feature Importance

Temporal trends in detection rate remained stable over the study period. A linear regression of detection rate against year was run to determine significant decline for each site. One site was declining at a significant rate, L455080, located in Rhode Island. Regression produced a slope = -0.036 / year, p value = $3.28e-08$, $R^2 = 0.911$ for site L455080.

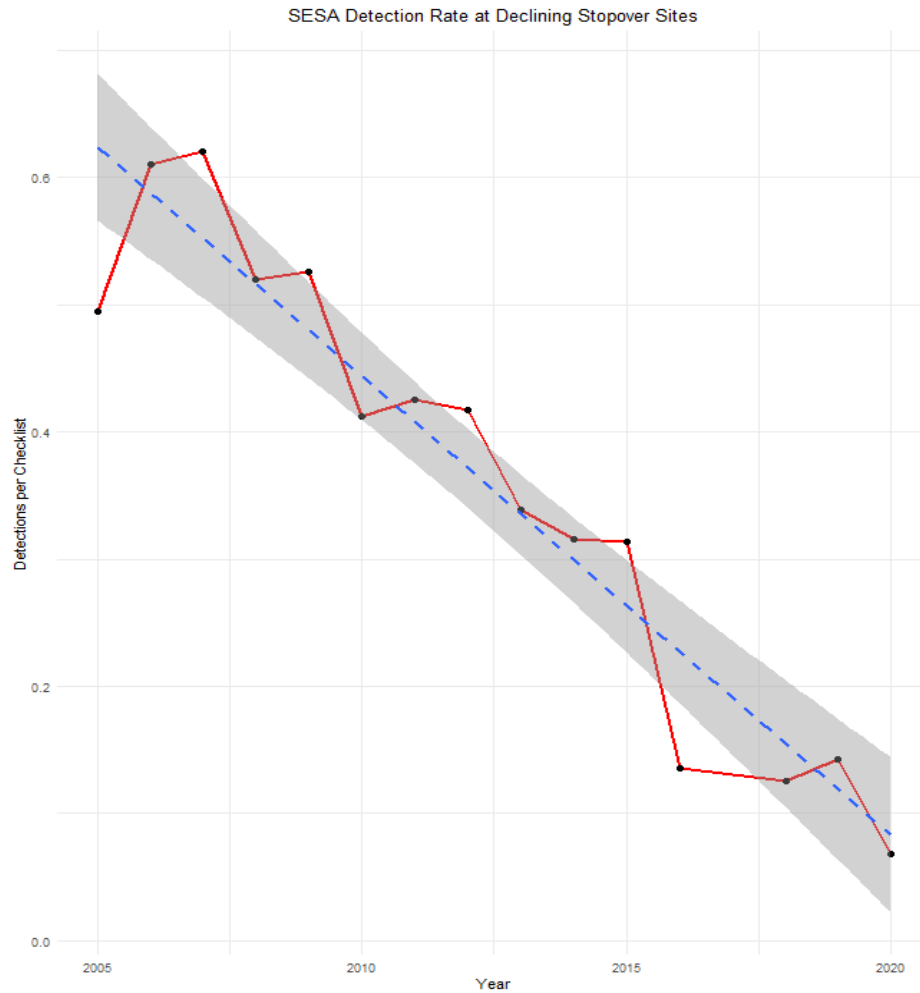


Figure 11. Detection rate trend at site L455080, Rhode Island (2005-2020)

Discussion

The hypothesis: SESA stopover site usage is predictable using land cover characteristics and sites that have experienced habitat degradation will be less likely to be used as stopover sites and therefore will have a lower chance of SESA being present, is partially supported by my analysis. Emergent herbaceous wetlands land cover class is a significant feature, with positive predictive power, meaning that wetlands are associated with high chance of SESA presence. Land cover

classes did not show the type of predictive power that physical land characteristics showed.

Elevation, distance to coast, and temperature were the dominant features, combining to about 65% and indicating that a site's characteristics mattered more than the type of habitat the site was labeled as. Habitat degradation did not have any influence on presence, which partially refuted the claim made in the hypothesis that stated a negative habitat change would aid in the prediction of absences.

When comparing the two models to one another, each had strengths and weaknesses. The LR captured only the simple linear relationships and struggled to capture interactions between features such as elevation and coastal distance that BRT was able to capitalize on. The lower recall score of the BRT model is a trade off that has to be considered based on the type of analysis being done. Higher recall is an important metric in model comparison when it comes to binary classification, and in this specific use case a higher recall is indicative of correctly labeling more true presence sites as such. If the model predicts an absence and it was in fact a presence site, this can lead to those sites going unmarked and in turn not being focused on for active conservation. Both models missed a substantial number of presence sites, (LR missed 55 and BRT missed 70) many of which were at developed areas of land. The model undervalued these classes, such as developed open space and low intensity development, even though SESA used this type of habitat.

The temporal findings derived from this analysis were meaningful and insightful. The finding that the detection rate is stable over the study period while the population is proving to be declining suggests that the top 500 stopover sites focused on are not the main source of the problem. These sites are desirable and therefore SESA returns to these sites over and over again. The lower quality sites that were not included in this analysis may be where the decline can be

seen more severely. The site that shows significant decline, L455080, is an anomaly. Objectively this site should not be showing significant decline, there was no habitat degradation, it has low elevation and is in close proximity to the coastline. This points to outside factors contributing to its decline. Factors that may be contributing, and should be considered as this analysis continues, are human disturbance, predator presence, food availability, smaller scale habitat aspects, and weather patterns.

There were many decisions made throughout this analysis, some strengthened the analysis and some were weaknesses or gaps that should be addressed in future work. Strengths included checklist based absence approach which is an ecologically valid approach and has been used by other previously published studies, improving upon the linear model with a nonlinear boosted regression tree model, and highlighting a species that is showing severe declining population trends over recent years. There are also some weaknesses within the project that need to be addressed. Ebird data can be biased due to its citizen science nature, there may be some misidentifications, and more populated or easily accessible areas may have a higher human presence and therefore a larger number of observations than more remote areas. Additionally, the NLCD land cover data may not capture fine scale habitat details that could prove to be essential for using land cover as a predictor. Spatial correlation was not formally tested. If a spatially close site ends up in both training and testing data sets, the evaluation metrics such as AUC may be inflated because the model saw similar data as the test set in the train set. Testing this using Morans I determines whether the residuals are spatially clustered or not, and therefore can determine if the model struggles when it comes to certain locations and therefore certain land classes. Finally, the BRT learning curves show that the model did not fully converge on the data,

this can potentially be improved by feeding the model more data (using more presence and absence sites) or including more features in the model.

This model has the potential to contribute meaningfully to SESA conservation efforts. Due to the model uncertainty, conservation decisions should be validated with collected field data. There is a possibility that when honing in on different habitat types, the model could perform differently.

The findings from this analysis can be used to evaluate areas in need of conservation or preservation, and expanding the data to evaluate more than the top 500 sites will allow for a larger scope to evaluate both high quality and low quality stopover sites. Citizen science data is collected by volunteers and not trained scientists, and because of this there are certain biases that must be taken into account, including location and identification standards.

A few key takeaways derived from this analysis include the undeniable fact that physical geography is far more impactful when it comes to prediction than land cover. Additionally, high quality stopover sites that were focused on in this analysis may be the buffer against population declines, when expanding the number of presence sites, more patterns relating to the decline may be uncovered.

Conclusion

The main objective of this analysis was to determine if the hypothesis was supported or rejected by the models. The main question to be answered was: Can SESA stopover sites be predicted using land cover characteristics? This hypothesis was partially supported by the findings.

Wetland cover was the most important land cover class, but compared to the physical land characteristics it was not as powerful of a predictor. Elevation, distance to the coastline, and

temperature were three of the most important variables used in predicting the presence of SESA. This leads to the idea that land cover class is not the most important variable to assess when determining what constitutes a probable SESA stopover site. In addition, citizen science can and does have biases and limitations, but despite this it can still be a useful data source for predictive modeling. Citizen science limitations can include but are not limited to incorrect identification of species as well as missed identification of species, especially small shorebirds such as SESA. When evaluating the population level declines that do not show up as seen by the steady detection rate, the sites evaluated play a large part. The top 500 sites that were looked at did not see a decline in detection rate, but this does not confirm that lesser used sites did not see significant decline.

Future research can include expanding the data to look at a larger volume of stopover sites, including a much greater percentage of the total unique sites identified within the dataset. Integrating future sea level rise scenarios to assess habitat vulnerability and predict which sites would increase in usage as vulnerable sites decrease. Occupancy modeling would aid in the biases that come with citizen science data by correcting for the fact that there are occasions where SESA are present at a site but not detected. Spatial cross validation of the models would improve their respective AUC values by ensuring that train and test data is not spatially similar therefore acting as data leakage. Finally, deploying this model to more migratory shorebird species that use stopover sites along the Atlantic coast of the United States would greatly increase the justification for conservation efforts of certain sites due to usage.

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